

AKSHAR AI
SUMMER INTERNSHIP REPORT
WEEK 3 — 08 JUNE 2026 – 13 JUNE 2026

Project Title:

Chest X-Ray Pneumonia Detection - Model Training

Field	Details
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Week 3 Objective

The objective of Week 3 was to understand and practice machine learning and deep learning model training techniques. The focus was on dataset preparation, train-test splitting, model building, training, evaluation, and performance comparison across multiple algorithms.

Task 1: Pneumonia Detection – Deep Learning Model Experiment (50 Images)

Dataset

- Dataset: Chest X-Ray Pneumonia Dataset
- Images Used: 50
- Classes:
 - Normal
 - Pneumonia

Data Preparation

- Selected 50 images from the original dataset.
- Resized images to a uniform size.
- Applied train-test split.
- Prepared image generators for model training.

Models Trained

- Custom CNN
- ResNet50
- DenseNet121

Learning Outcomes

- Understood CNN architecture and convolution operations.
- Learned transfer learning using pretrained models.
- Practiced model compilation, training, validation, and evaluation.
- Compared performance of multiple deep learning architectures.

Key Observation

DenseNet121 achieved the best performance among the tested deep learning models due to efficient feature reuse and dense connectivity.

Task 2: Pneumonia Detection – Machine Learning vs Deep Learning Comparison (50 Images)

Objective

To understand the difference between traditional machine learning algorithms and deep learning models for image classification.

Machine Learning Models

1. Logistic Regression
2. Support Vector Machine (SVM)
3. Random Forest

Deep Learning Models

1. CNN
2. ResNet50
3. DenseNet121

Process

- Images were resized and converted into numerical features.
- For ML models, images were flattened into feature vectors.
- For DL models, images were used directly for feature extraction and learning.

Learning Outcomes

- Learned feature extraction and image flattening.
- Understood the limitations of traditional machine learning for image data.
- Learned how deep learning automatically extracts meaningful image features.

Key Observation

Deep learning models outperformed traditional machine learning models because they preserve spatial information and learn image features automatically.

Task 3: Emotion Detection Using Machine Learning Models

Dataset

- Dataset: Happy and Sad Faces Dataset
- Total Images: 180
- Happy Images: 90
- Sad Images: 90

Exploratory Data Analysis (EDA)

Performed:

- Dataset overview
- Class distribution analysis
- Sample image visualization
- Image dimension analysis
- Data preprocessing and resizing

Data Preprocessing

- Converted images to grayscale.
- Resized all images to 64×64 pixels.
- Flattened image matrices into feature vectors.
- Applied 80:20 train-test split.

Models Trained

1. Logistic Regression
2. Support Vector Machine (SVM)
3. Random Forest

Results

Model Accuracy

Logistic Regression - 63.89%

SVM - 72.22%

Random Forest - 58.33%

Key Observation

SVM achieved the highest accuracy among the machine learning models. The experiment demonstrated that SVM performs effectively on small image datasets with high-dimensional feature spaces.

Skills Learned During Week 3

- Train-Test Splitting
- Image Preprocessing
- Feature Extraction
- Logistic Regression
- Support Vector Machine (SVM)
- Random Forest
- Convolutional Neural Networks (CNN)
- Transfer Learning
- ResNet50
- DenseNet121
- Model Evaluation
- Accuracy Comparison
- Machine Learning vs Deep Learning Analysis
- Exploratory Data Analysis (EDA)

Conclusion

Week 3 focused on practical model training and performance evaluation. Multiple machine learning and deep learning models were trained on image datasets to understand their strengths and limitations. Through experimentation with pneumonia and emotion detection datasets, valuable insights were gained into image preprocessing, feature extraction, transfer learning, and model comparison techniques. These activities strengthened the understanding of model development workflows and prepared the foundation for advanced

topics such as data augmentation, hyperparameter tuning, explainable AI, and deployment.